Preregistration: Experimental Block V

Reflective Self-Modelling

Theory of Everything Research Programme

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1. Status and purpose

This record freezes the first confirmatory study for Experimental Block V. No Block V confirmatory seed or confirmatory-only task variant has been executed at the time of freezing. Development-only simulations were used to validate the software, expose defects, strengthen adversarial controls and choose prospective thresholds; those outcomes are excluded from confirmatory evidence. Block V distinguishes successful control, self-prediction and causally used self-related representations.

2. Architecture classes

The frozen comparison uses S0 no explicit self-monitor, S1 passive self-monitor, S2 active self-monitor, S3 shuffled/distribution-matched self-monitor, S4 capacity-matched non-self control, H_ERROR task-specific recent-error bypass and H_EXTERNAL externally supplied calibrated-confidence bypass. H_EXTERNAL is intentionally strong and may defeat necessity claims.

3. Held-out confirmatory structure

Confirmatory evaluation uses task variants marked unexecuted during development. The frozen set includes held-out decision and verification schedules, an unannounced latent-impairment switch, fixed misinformation and correction conditions, an exact-multiset correspondence-broken shuffled monitor, a novel capability boundary, a self-versus-world uncertainty test and a bypass suite.

4. Primary hypotheses

H5.1 tests selective benefit from active rather than passive self-monitoring. H5.2 tests recovery after unannounced internal impairment. H5.3 tests direct misinformation and correction/rescue. H5.4 tests episode-specific monitor correspondence against exact-distribution shuffled control. H5.5 separates internally generated capability change from external-world uncertainty. H5.6 tests whether simpler bypasses reproduce the decisive functions.

5. Seeds and execution gate

Twenty unique confirmatory seeds are frozen in BLOCK5_CONFIRMATORY_SEEDS_V1.txt. The seed-independence audit passes. The held-out-variant audit records all confirmatory variants as unexecuted during development. The experiment registry remains FROZEN_PREREGISTRATION_READY_UNRUN and confirmatory execution is refused until a public OSF preregistration is recorded.

6. Decision criteria

H5.1 RETAIN requires S2-S1 mean accuracy >=0.12 in both self-relevant families, >=18/20 seeds favouring S2 and <=0.05 absolute difference on matched irrelevant controls. H5.2 RETAIN requires >=0.15 post-impairment S2-S1 accuracy in both families, median detection <=3 episodes, self-estimate decrease >=0.10 and >=18/20 seed consistency. H5.3 RETAIN requires overconfidence-related accuracy loss >=0.15 in both families plus corresponding control suppression, correction rescue >=0.15 and action-allocation restoration within 0.05. H5.4 RETAIN requires S2-S3 post-impairment accuracy >=0.15 in decision and >=0.10 in verification, with exact marginal monitor distribution preserved. H5.5 RETAIN requires self-monitor MAE >=0.05 better than world-only uncertainty, self-estimate drop >=0.10 and world-only change <=0.02. H5.6 RETAIN requires S2 to exceed every bypass by >=0.10 on both families; REDUCE is explicitly triggered if H_EXTERNAL remains within 0.05 of S2 on both families while reproducing the downstream control benefit.

7. Analysis and reporting

Independent seeds are the primary replicate unit. Calibration is analysed separately from behavioural benefit and is measured on unassisted outcomes before monitor-triggered intervention changes correctness. Threshold decisions use the frozen manifest rules. No scalar self-awareness score is constructed. Successful bypasses narrow or reduce necessity claims rather than being treated as anomalies.

8. Development disclosure

Development-only results established workable effects for active monitoring, impairment recovery and misinformation/correction. The S2-S1 development accuracy gains were 0.248 in the decision family and 0.157 in the verification family. H_EXTERNAL matched S2 closely in both development families, motivating an explicit H5.6 REDUCE route. A calibration/control conflation found during hardening was corrected before freezing.

9. Scope and limitations

A positive Block V result can support calibrated self-monitoring and causally used self-related representations within the registered artificial task families. It cannot establish subjective experience, phenomenal awareness, sentience, human-like selfhood or conscious introspection. Paper III ends the principal hierarchy at reflective self-modelling because Paper I identifies no validated bridge from reflective self-modelled intelligence to phenomenal experience.